Enterprise-Inspired Network Topology Project

1. Introduction

The project shows the configuration of a multi-site enterprise network where two geographically separated locations communicate through the Internet. The network is designed to simulate a real world corporate infrastructure where Headquarters (Hubs) connect with branch offices (Spokes).

The two sites communicate over the public Internet using a site-to-site GRE (Generic Routing Encapsulation) tunnel, which allows internal devices to communicate across a public network.

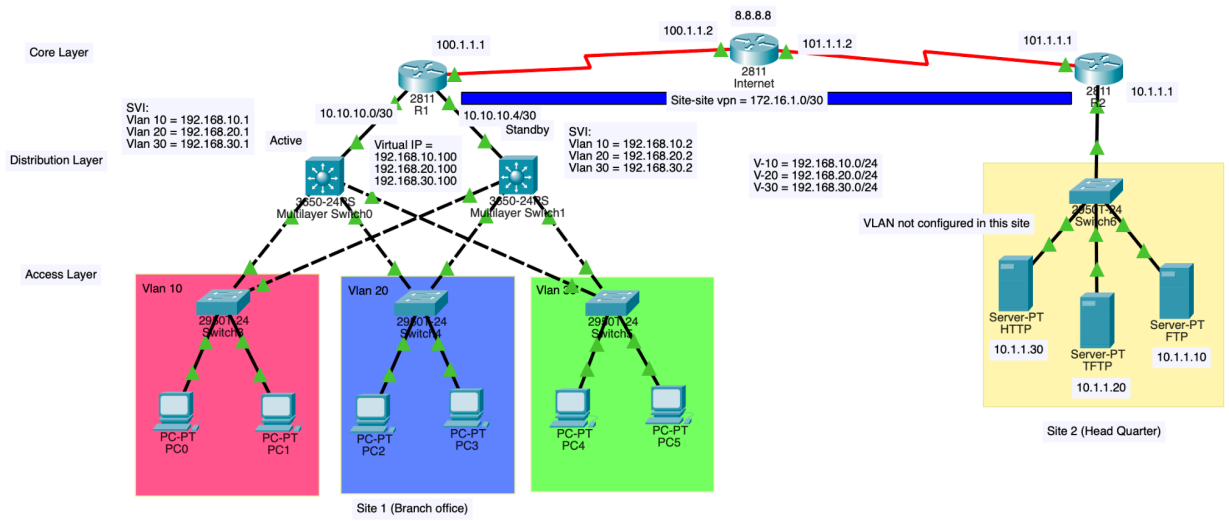
In addition to a VPN connection, the network includes various networkin protocols and technologies:

- VLANs to segment users into sub networks
- Trunk links to carry VLAN traffic between the switches
- Switch Virtual Interfaces (SVIs) for layer 3 routing of VLAN traffic
- HSRP (Hot Standby Router Protocol) for gateway redundancy
- OSPF (Open Shortest Path First) for reachability routing
- NAT/PAT (Network Address Translation) to allow internal devices to connect to the internet
- GRE VPN tunnel to connect the branch office with the headquarters.

Goal:

- Devices at both sites can access the Internet
- Branch office devices can access servers in the HQ
- The network should still continue to operate even if one distribution layer switch fails due to HSRP redundancy

2. Network Topology overview



The topology is divided into two sites.

Headquarters (Site 2)

The headquarters contains:

- A router
- An access layer switch
- Three servers:
    1. HTTP server
    2. FTP Server
    3. TFTP Server

The router connects the headquarters to the Internet and also participates in the GRE VPN tunnel with the branch office.

Branch office (Site 1)

The branch office uses a three-layer hierarchical architecture:

- Access Layer
  Connects to end devices such as PCs.
  Each access layer switch contains devices belonging to a specific VLAN.
- Distribution Layer
  Performs:
    1. Inter-VLAN routing
    2. HSRP redundancy
    3. Connects to the Core Layer

- Core Layer
  A router that connects the internal network to the ISP.

3. IP Addressing Plan

VLAN Networks:

VLAN	Network
VLAN 10	192.168.10.0/24
VLAN 20	192.168.20.0/24
VLAN 30	192.168.30.0/24

Each VLAN has its own subnet with a broadcast domain.

HSRP Virtual Gateways:

VLAN	Virtual Gateway
VLAN 10	192.168.10.100
VLAN 20	192.168.20.100
VLAN 30	192.168.30.100

PCs will use the virtual IP address as their default gateway.
HSRP ensures that if one switch fails, another switch automatically becomes the default gateway, so we have this Active and passive relationship.

Core Router Links:

Links	Network
Gig 1/0/4 DSW1 <-> Fa 0/0 R1	10.10.10.0/30
Gig 1/0/4 DSW2 <-> Fa 0/1 R1	10.10.10.4/30

Each router interface has its own broadcast domain

Internet Connections:

Links	Network
HQ Router <-> ISP	101.1.1.0/30
Branch router <-> ISP	100.1.1.0/30

Headquarters LAN:

Device	IP Address
Core Router	10.1.1.1
HTTP Server	10.1.1.30
FTP Server	10.1.1.10
TFTP Server	10.1.1.20

4. Setup

Edge Device Config
Before configuring the network infrastructure, each PC must have an IP address so it can communicate with the network.

PC1

IP Address: 192.168.10.10 (VLAN 10 IP)
Subnet Mask: 255.255.255.0
Gateway IP: 192.168.10.100 (Virtual IP)

If the active switch fails, another switch takes over without any change on the PC. This is why the gateway IP is the HSRP virtual router.

*Same config for the rest of the PCs

Servers

IP Address: 10.1.1.30
Subnet Mask: 255.255.255.0
Gateway IP: 10.1.1.1

*Same config for the rest of the servers

5. Access Layer Switch Configuration

The first networking devices configured are the access switches.

These switches connect directly to edge devices. Each switch is assigned a specific VLAN.

Step 1 (Create VLAN)
Step 2 (Assign interfaces to VLAN)

Commands:
Switch 3

```
enable
configure
vlan 10
Interface r fa 0/1 - 2
switchport mode access
switchport access vlan 10
```

Explanation:

- VLANs divide a physical network into multiple logical networks
- Devices in different VLANs cannot communicate without routing
- Switchport mode access forces the layer 2 interface to have one VLAN
- Switchport access vlan 10 assigns that interface to VLAN 10

*Same config for VLAN 20 and VLAN 30

6. Configure Trunk Links

Distributed layer switches must receive traffic from multiple VLANs

We need trunk links.

Commands:
Switch 3

```
interface r Gig 1/0/1 - 3
switchport mode dynamic desirable
```

Explanation:

- Trunk interfaces allow VLAN traffic to be routed to the layer 3 switches using DTP (Dynamic Trunking Protocol)

7. Configure Switch Virtual Interfaces (SVIs)

SVIs allow the switch to function as a layer 3 device and route between VLANs

Commands:
DSW 1

```
interface vlan10
ip address 192.168.10.1 255.255.255.0
exit
```

*The same for vlan 20 and vlan 30

DSW 2

```
interface vlan10
ip address 192.168.10.2 255.255.255.0
exit
```

*The same for vlan 20 and vlan 30

Explanation:
- This IP address becomes the default gateway candidate for devices in that VLAN
- Its important that the IP address for each SVI interface on both switches have the same IP!

8. Configure HSRP (Gateway Redundancy)

HSRP allows two switches to share one virtual gateway address.
One switch is active and forwards traffic.
The other switch is standby and takes over if the active device fails.

DSW 1 (Active)

```
interface vlan10
standby 10 ip 192.168.10.100
standby 10 priority 110
standby 10 preempt
```

Explanation:

- Standby 10 -> This HSRP group number
- Ip -> Defines virtual gateway ip
- Priority 110 -> Makes the switch active
- Preempt -> Switch reclaims active status once back online
- My design choice is configuring DSW1 to be active for VLAN 10, 20 and 30.

DSW 2 (Standby)

```
interface vlan10
standby 10 ip 192.168.10.100
```

Remember that the default priority of the switches are 100, so now DSW 2 becomes standby switch.

9. Configure Layer 3 links to Router

By default, switchports are layer 2 interfaces.

To allow routing, they must be converted to layer 3 interfaces.

Commands:
DSW1

```
interface Gig 1/0/4
no switchport
ip address 10.10.10.1 255.255.255.252
no shutdown
```

*Do the same with DSW2 but with ip address 10.10.10.5/30

Explanation:

- By default router interfaces are disabled to ensure saftey until network admin is ready to enable. If interface shows administratively down run the no shutdown command.

10. Configure OSPF Routing

OSPF is used so that devices can share reachability info to learn routes.
Without routing protocols, administrators need to configure static routes manually.

Commands:
DSW1

Enable OSPF

```
router ospf 1
network 10.10.10.1 0.0.0.0 area 0 //configure ospf on interface Gig 1/0/4
network 192.168.10.0 0.0.0.255 area 0 //configure ospf on vlans
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
```

*Do the same from DSW 2

Explanation:

- 1 is the ospf process ID

Commands:
R1

```
router ospf 1
network 10.10.10.2 0.0.0.0 //configure ospf on interface on fa 0/0
```

Now we remove neigbor redundancy between DSW 1 and DSW 2. We only want a neighbor between the router and DSW not between DSW 1 and DSW 2.

Commands:
DSW 1

```
router ospf 1
passive-interface vlan 10
passive-interface vlan 20
passive-interface vlan 30
```

```
DSW2#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.1.1        1   FULL/BDR        00:00:30    10.10.10.6      GigabitEthernet1/0/4
DSW2#
```

```
DSW1#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
172.16.1.1        1   FULL/BDR        00:00:30    10.10.10.2      GigabitEthernet1/0/4
```

Explanation:

- This tells OSPF to advertise the VLAN prefix to the router.
- Ensure that the interface IP between router and distributed layer are being advertised.

11. Configure WAN links to internet

Commands:

R1

```
enable
configure
interface se0/2/0
ip address 100.1.1.1 255.255.255.252
no shutdown
```

*Do the same for se0/0/0 and se0/2/0 for other WAN link with R2

Internet

```
enable
configure
interface se 0/2/0
ip address 100.1.1.2 255.255.255.252
no shutdown
exit
interface l1 //Create lookback
ip address 8.8.8.8 255.255.255.255
```

Explanation:

- Note serial connections are used for WAN interfaces
- A loopback interface is a virtual interface that is always active unless manually disabled. It is often used to simulate networks or test connectivity. In this case devices communicate with 8.8.8.8 to simulate an internet connection.

12. Configure NAT for Internet Access

Private IP addresses in the range of 192.168.x.x cannot travel across the Internet.

NAT translates them into public IP addresses.

Step1 (Create ACL):

Commands:
R1

```
access list 10 permit 192.168.0.0 0.0.255.255
```

Explanation:

- Identifies which range of internal addresses should be translated

Step 2 (Enable PAT):
Commands:
R1

```
ip nat inside source list 10 interface se0/2/0 overload
```

Explanation:

- PAT allows many internal devices to share one public IP address

Step 3 (Define NAT Interfaces):

```
Interface fa0/0
Ip nat inside

Interface fa0/1
Ip nat inside

Interface se0/2/0
Ip nat outside

Ip route 0.0.0.0 0.0.0.0 100.1.1.2 //Default route between core and
Internet
```

*Do the same with R2 but with ip range 10.1.1.0/24; fa0/0 NAT inside and se0/2/0 as NAT outside. Default route must be created between core router and internet router with R2 with 101.1.1.2

Explanation:

- Set interfaces to internal or external for NAT config; otherwise, translation is not possible

13. Headquarter router configuration

The branch router must also allow servers to access the Internet. So NAT configuration must be done on router 2.

Commands:

R2

```
interface fa0/0
ip address 10.1.1.1 255.255.255.0
ip nat inside
```

14. Configure GRE Tunnel

The GRE (site-to-site) tunnel allows two sites to communicate with each other over the internet.

Commands:
R2

```
interface tunnel1
tunnel source se0/2/0
tunnel destination 100.1.1.1
ip address 172.16.1.2 255.255.255.252
```

R1

```
interface tunnel1
tunnel source se0/2/0
tunnel destination 101.1.1.1
ip address 172.16.1.1 255.255.255.252
```

15. Run OSPF over the Tunnel

We want the sites to learn each other's networks.

R1

```
router ospf 1
network 172.16.1.1 0.0.0.0 area 0
```

R2

```
router ospf 1
network 172.16.1.2 0.0.0.0 area 0
```

16. Verification

Test Internet access:

```
ping 8.8.8.8
```

Test VPN connection:

```
ping 10.1.1.30
```